

Probability, Statistics & Data Analysis Problem Sheets 2024

Contents

1.	Descriptive Statistics and Visualisation	3
1.1.	Descriptive Statistics (to be done by hand).....	3
1.2.	Descriptive Statistics (to be done with MATLAB).....	3
1.3.	Plotting and Analysis using MATLAB.....	4
2.	Probability Theory	5
2.1.	Visa or Mastercard?	5
2.2.	Blue and Red Dice	5
2.3.	Marbles.....	5
2.4.	Defective Bolts	5
2.5.	Student class distributions.....	5
3.	Discrete Random Variables.....	6
3.1.	Expected values, variance and CDF.....	6
3.2.	Buses and students	6
3.3.	Tennis rackets	6
3.4.	Nursing home	6
3.5.	Earthquakes.....	6
3.6.	Metal detector	7
3.7.	Poisson Pores	7
4.	Continuous and Joint Random Variables	9
4.1.	Uniform PDF	9
4.2.	Archery	9
4.3.	Foraging ants	9
4.4.	Insurance Company	9
4.5.	Bank facilities	10
4.6.	Electric Cable.....	10
5.	Normal Distribution, Statistics and Intervals.....	11
5.1.	Blood chloride concentration.....	11
5.2.	iPerfusion Data - Part 1.....	11
5.3.	MATLAB Simulation of CLT	12
6.	NHST and z-tests.....	14
6.1.	Environmental Agency.....	14
6.2.	Soil Specimens – Part 1	14
6.3.	Synthetic Biodiesel – Part 1	15
6.4.	iPerfusion Data – Part 2	15
7.	T-tests and assumptions.....	17
7.1.	Soil Specimens – Part 2	17

7.2.	Synthetic Biodiesel – Part 2	18
7.3.	Happyculture vs O-BeeHive	18
8.	Non-Parametric Tests and Regression	20
8.1.	Ventilator Battery	20
8.2.	Soil Specimens – Part 3	20
8.3.	Synthetic Biodiesel – Part 3	21
8.4.	Crude oil imports vs chicken consumption	21
8.5.	Heart rate vs life expectancy	22
8.6.	Hip and knee forces	22
9.	Multiple Comparisons	24
9.1.	E. coli doubling rate	24
9.2.	Titanium bone implants: parametric	24
9.3.	Titanium bone implants: non-parametric	25
9.4.	Intraocular pressure medication	25
9.5.	Corrosion fatigue	26
Bonus -	ANOVA with MATLAB	27
9.6.	Parametric titanium bone implants with MATLAB	27
9.7.	Non-parametric titanium bone implants with MATLAB	27
9.8.	Intraocular pressure medication with MATLAB	27
9.9.	Corrosion fatigue with MATLAB	27

1. Descriptive Statistics and Visualisation

1.1. Descriptive Statistics (to be done by hand)

$x = [45, 4, 6, 9, 28, 15, 16, 22, 29, 31, 40, 65, 7]$

Calculate the following measures of location (to the nearest integer)

- a) Mean
- b) Median

Calculate the following measures of spread (to the nearest integer)

- c) Standard deviation
- d) Median absolute deviation with $k = 1.4826$
- e) Inter-quartile range
- f) Reflection: why did I only ask for an integer, rather than more decimal places?

[Answers: 24, 22, 18, 19, 22]

1.2. Descriptive Statistics (to be done with MATLAB)

Load data 'Exercises1_data.mat'. In this question, we will use data set A.

For each measure below, write `MATLAB` functions that calculate the statistics manually and compare your values to the in-built `MATLAB` functions (which you will have to find). Some functions you may wish to use are given in parentheses (the first time they are required: they may also be needed subsequently).

For the reasons above, report your answers to the nearest integer using `round`.

- a) Mean (`sum`, `numel`)
- b) Median (`sort`, `if/else`)
- c) Standard deviation (`sqrt`)
- d) Median absolute deviation with $k = 1.4826$ (`abs`). For the in-built function, check help for the correct input arguments

[Answers: 21, 21, 4, 3]

For percentiles, do not round as we are dealing with specific values. You'll find the answers are slightly different with our definition and the `MATLAB` code due to the interpolation used by the latter.

- e) 25th percentile, $\eta(0.25)$ (`ceil`, `prctile`)

[Answers: 18.07, 18.04]

1.3. Plotting and Analysis using MATLAB

Load data 'Exercises1_data.mat'. This contains three arrays of a variable of different sizes A, B and C. The aim here is to visualise the data. Let us imagine that the numbers represent the amount of time spent on the Ed discussion board for this class per week, from three different cohorts.

Use the MATLAB code that was used to generate data in the lecture to guide you. For part b, you will need the functions `plotSpread.m` and `distributionPlot.m` as well.

The purpose of this is partly to familiarise yourself with MATLAB, but more importantly to think about how best to show the information.

For each figure, make sure to label the axes appropriately and to use the same axis for each data set.

a) Histograms

Use a 1x4 subplot with the following format

All three histograms on top of each other	Histogram for A	Histogram for B	Histogram for C
---	-----------------	-----------------	-----------------

- Create a figure and generate the histograms as in the table using default bin edges and count (just use the `histogram` command)
- Create a new figure and generate the histograms of the data using bin edges of -10, -8, -6, ..., 36, 38, 40 and probability normalisation for the y-axis
 - Make sure the y-axis ranges of all figures are the same
 - Optional: Choose your own colour scheme (using RGB or [built-in colour maps](#))
- Consider the plots
 - Which visualisation (i or ii) gives you a better understanding of the data?
 - What can you say about the shapes of the distributions?
 - How do you think the mean and median will compare for data set A as compared to data set B
 - What is the best average to use for data set C?

b) Dot plots, Beeswarm Plots, Violin plots

Use a 2x4 subplot with the following format

Boxplot	Box plot + Scatter plot
Beeswarm plot	Violin plot

- Create each of the figure types.
- Consider the plots
 - Which visualisation gives you a better understanding of the data?
 - What might you add to the figures?

2. Probability Theory

2.1. Visa or Mastercard?

Consider a randomly selected student. Let A denote the event that the selected individual has a Visa card and B be the event that they have a MasterCard. Suppose that $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cap B) = 0.25$.

- Compute the probability that the selected individual has at least one of the two types of cards, i.e., $P(A \cup B)$.
- What is the probability that the selected individual has neither type of card?
- Describe, in terms of A and B , the event that the selected student has a Visa card but not a MasterCard, and then calculate the probability of that event.

[Answers: 0.65, 0.35, 0.25]

2.2. Blue and Red Dice

Consider rolling two 2 dice, one blue and one red.

- What is the probability that the number on the red die is greater than the number on the blue die?
- What is the probability that the number on the blue die is greater or equal to the number on the red die?
- What is the probability of getting the same number on the two dice?

Hint: draw a table containing all the possible outcomes in the space of the joint experiments

[Answers: 0.42, 0.58, 0.17]

2.3. Marbles

You have a bag containing 3 green marbles and 5 blue marbles. When selecting 3 marbles, what is the probability of getting 1 green and 2 blue marbles if

- each marble is replaced after selection
- each marble is set aside after selection

[Answers: 0.44, 0.54]

2.4. Defective Bolts

A factory production line is manufacturing bolts using three machines A, B and C. Of the total output, machine A is responsible for 25%, machine B for 35% and machine C for the rest. It is known from previous experience with the machines that 5% of the output from machine A is defective, 4% from machine B and 2% from machine C. A bolt is chosen at random from the production line and found to be defective. Which machine is it most likely the bolt came from?

[Answers: Machine B]

2.5. Student class distributions

There are 180 students in a 3rd year Probability and Statistics class. For study groups, they are divided randomly into 3 classes of 60 pupils each. Alice, Bob and Caroline are all in the school year.

- What is the probability that Alice and Bob end up in the same class?
- What is the probability that Alice, Bob and Caroline all end up in different classes?

[Answers: ~0.33, ~0.23]

3. Discrete Random Variables

Questions 1-6 should be done by hand

3.1. Expected values, variance and CDF

Let X be an RV that can take the following values: $X=\{0,1,2,3\}$ with probabilities $P(0)=0.1$, $P(1)=0.5$, $P(2)=0.3$, $P(3)=0.1$

Questions:

- Calculate $E(X)$ and $V(X)$
- Calculate the CDF of X
- Hence find the probability that X is within one standard deviation (square root of variance) of its mean (expected value)

[Answers: 0.64, ~ , 0.8]

3.2. Buses and students

A total of four buses carrying 148 students arrive at a football stadium. The buses carry, respectively, 40, 33, 25 and 50 students.

- One of the students is randomly selected. Let X denote the number of students that were on the bus carrying the randomly selected student.
 - One of the four bus drivers is also randomly selected. Let Y denote the number of students on their bus.
- Intuitively, which of $E(X)$ or $E(Y)$ is larger? Why?
 - Compute $E(X)$ and $E(Y)$.

[Answers: ~ , 39.3 & 37.0]

3.3. Tennis rackets

A particular type of tennis racket comes in a midsize version and an oversize version. Sixty percent of all costumers at a certain store want the oversize version.

- Among the ten randomly selected customers who want this type of racket, what is the probability that at least seven want the oversize version?
- The store currently has eight rackets of each version. What is the probability that all of the next ten customers who want this racket can get the version they want from the current stock?

[Answers: 0.38, 0.95]

3.4. Nursing home

A nursing home finds that a patients have strokes at an average rate of L strokes per month (assume L is small and treat a month as 4 weeks). What is the probability of 2 strokes next week as a function of L ? Hence find the probability of 2 strokes next week if $L =$ one stroke per month.

[Answer: 0.024]

3.5. Earthquakes

Huge earthquakes hit LA at average rate of one in a century. What is the probability of NOT getting a huge earthquake in the next 10 years?

[Answer: 0.91]

3.6. Metal detector

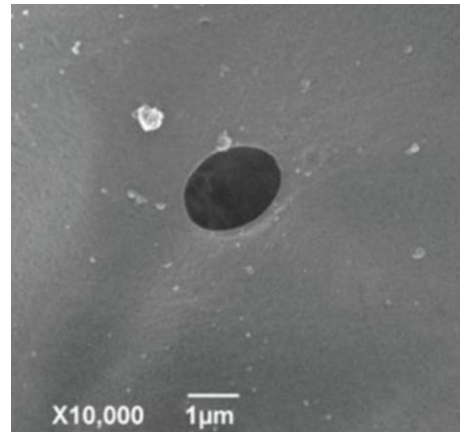
Of the people passing through an airport metal detector, 0.1% activate it. Let X be the number of individuals from a randomly selected group of 600 who activate the detector. A delay will be caused if 2 or more people are stopped. What is the probability that this happens?

[Answer: 0.12]

3.7. Poisson Pores

This question uses MATLAB

This question is inspired by the paper [Altered mechanobiology of Schlemm's canal endothelial cells in glaucoma](#). Part of this study involved counting ~micron size pores that form in the drainage tissues of the eye and play a key role in the regulation of intraocular pressure. The experiments involved fixing tissues from healthy eyes and those with glaucoma, and taking Scanning Electron Microscopy images of the endothelial cells from Schlemm's Canal. Each image was 200µm by 200µm and contained a small number of pores. 150 images from healthy eyes and 150 images from glaucomatous eyes were investigated and the number of pores were counted.



The data for this question (based on real data) can be generated yourself. Use the following code

```
x=0:20;
Nh=round(150*poisspdf(x,8));
Ng=round(150*poisspdf(x,2));
```

You now have the following variables in the workspace

- x – the possible values of the random variable X that represents the number of pores per image
- N – the number of images containing x pores for each value of x
 - N_h is for healthy eyes
 - N_g is for glaucomatous eyes

To those who are not familiar with MATLAB, this might be a bit of a challenge before you see the solutions, but look at the code from the lecture and try without the solutions first

Convert the N values into probabilities (i.e. calculate the PMF)

Create a figure with two subplots. On the first subplot, plot the PMF for both samples on a single axis with a colour scheme of your choice. I suggest using a `bar` plot (this is the only time I will do so!).

You can try to make it look good (transparency helps)

Calculate the CDF using the `cumsum` function for each sample

On the second subplot, plot the CDF for both samples on a single axis with a colour scheme of your choice. I suggest using `plot` to create lines, as bars may be less clear. Make it look good!

Calculate the Expected Value of each sample (h and g) using

a `for` loop

the `sum` function, making use of element wise multiplication (`.*`), where the dot results in a multiplication of each element at a time, rather than a matrix multiplication.

They should be the same

[Answer: the expected values should be ~8 and ~2]

Calculate the Standard Deviation of each sample (h and g) using

a `for` loop using $V(X) = \sum (x - E(X))^2 p(x)$

using the `sum` function using $V(X) = E(X^2) - [E(X)]^2$ and making use of element wise multiplication (`.` `*`)

They should be the same

[Answer: the variances should be ~8 and ~2]

Ponder

What do you notice about the expected value and variances.

Can you identify why the values aren't exactly what you might have expected?

Do you think the number of pores is meaningfully different in glaucoma patients?

4. Continuous and Joint Random Variables

All questions can be done by hand. For questions 3 and 4, MATLABs symbolic toolbox is useful for checking your answers

4.1. Uniform PDF

If X is a continuous uniform RV, defined according to

$$f(x) = \begin{cases} k & 2 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- Find k and hence sketch $f(x)$
- Calculate the probability $P(3 \leq X \leq 3.5)$
- Discuss the difference between the probabilities $P(3 < X < 3.5)$ and $P(3 \leq X \leq 3.5)$

[Answers: 0.5, 0.25, ~]

4.2. Archery

This question is about archery. Let X be the distance in inches of the location of the arrow from the centre of the target. X has the following PDF:

$$f(x) = \begin{cases} \frac{1}{10} - \frac{x}{200}, & 0 \leq x \leq 20 \\ 0, & \text{otherwise} \end{cases}$$

- Draw a graph of $f(x)$
- Calculate the CDF
- Using the CDF, calculate the probability that an arrow will be between 5 and 10 inches of the target centre

[Answers: ~, ~, 0.3125]

4.3. Foraging ants

Ants leave their colony from a single exit to forage for food. As you move away from the colony, the probability of finding an ant decreases according to $f(x) = ae^{-bx}$ where the RV X is the distance from the colony exit point, and hence $0 \leq x < \infty$.

- Assuming that this is a valid PDF, find the relationship between a and b and state $f(x)$ as a function of b and x only
- Find the median of X as a function of b
- Find the expected value of X
- Find the variance of X . This one gets a little heavy on the algebra. Hint: it's easiest to multiply it out first, and you can then use the solution of the integral by parts from part (c) a couple of times

[Answers: $a=b$, $-\ln(0.5)/b$, $1/b$, $1/b^2$]

4.4. Insurance Company

An insurance company offers both phone and laptop insurance. They want to know how much they make from each, and whether getting one makes one more likely (or less likely) to get the other. They gathered the following data

		Amount for phone insurance (£)		
		0	7	10
Amount for laptop insurance (£)	0	20	24	16
	12	27	42	13
	15	19	11	18

For the following calculations you may use fractions, or you can round your numbers to 3 decimal places without loss of accuracy

- Generate the Joint Probability Table, including the marginal PMFs
- Find the expected amount to make on phones alone from each customer
- Find the expected amount to make on laptops alone from each customer
- Are X and Y independent?
- Calculate the correlation coefficient. What can you conclude?

[Answers: ~, 5.305, 8.967, ~, -0.022]

4.5. Bank facilities

A bank operates a drive-up facility and a walk-up window. Let X be the proportion of time that the drive-up facility is used, and Y the proportion that the walk-up window is used. Suppose that the probability density function $f(x, y)$ is given by

$$f(x, y) = \begin{cases} \frac{6}{5}(x + y^2) & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- Verify that this is a valid joint PDF.
- Find the marginal probability distribution functions $f_X(x)$ and $f_Y(y)$
- Hence identify whether X and Y are independent. Check your answer manually and/or with the `subs` (substitute) function by investigating (X, Y) equal to $(0.5, 0.5)$ and $(0.3, 0.3)$

Note: for the rest of this question, there is really quite a lot of algebra, so it can be a little time consuming. You may want to do it all with `MATLAB` and save the manual algebra for a rainy day. Of course, the best practice would be to solve by hand and with `MATLAB` and thereby confirm that you have both techniques nailed!

- Calculate the covariance between X and Y . You can do this using the complete formula given in the lecture notes, or using the shortcut $cov(x, y) = E(XY) - E(X)E(Y)$. Or ideally, use both and show that they are the same!

The functions `subs`, `expand` and `simplify` are useful if you go down the `MATLAB` route

- Hence calculate the population correlation coefficient

[Answer: ~, ~, ~, -0.01, -0.131]

4.6. Electric Cable

X denotes the diameter of an electric cable and Y denotes the diameter of the ceramic mold that is used to make the cable. Both X and Y are scaled so that they have a range between 0 and 1. The joint PDF is given by

$$f(x, y) = \begin{cases} 1/y & 0 < x < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the probability that $P(X + Y < 2/3)$

[Answer: $2/3 \ln(2)$]

5. Normal Distribution, Statistics and Intervals

5.1. Blood chloride concentration

Long term studies of blood chloride concentration (X) in the population demonstrate that it is appropriate to characterise X as normally distributed with a mean 101 mmol/l and standard deviation 3 mmol/l, $X \sim N(101, 3^2)$.

- a) What is the probability that for a given individual $x > 104 \text{ mmol/l}$
[Answer: 0.159]
- b) What is the probability that for a given individual, x is more than 2 standard deviations away from the mean and does it depend on the standard deviation?
[Answer: 0.046]
- c) For a sample of 10 individuals, the sample average $\bar{x}$ is to be calculated. What will the 99% margin of error be?
[Answer: 2.44]
- d) How many observations would be required in a given sample in order to only expect a 5% chance of $\bar{x}$ being less than 100 mmol/l or greater than 102 mmol/l
[Answer: 35]
- e) A drug is used to decrease the blood chloride concentration. It is assumed that this will change the mean value, but not the standard deviation. A sample with 10 observations is used to calculate a sample mean of $\bar{x} = 96.0$. Using the 95% prediction interval, could it be considered likely that the next observation is larger than the population mean of the untreated population?
[Answer: yes]
- f) A sceptical individual suggests that perhaps the drug actually hadn't worked, and the sample mean of the treated population was just observed due to random sampling. If this was true, is the observed likely (using $\alpha = 0.05$)?
[Answer: no]
- g) The sceptic presses further, certain that more information can be gleaned from the analysis.. "What is the probability of seeing this outcome or an even smaller value?"
[Answer: $p < 0.0002$]
- h) *Reflections.* Consider/discuss the following
 - Under what circumstances PI is similar to $\bar{x} \pm 2s$, and why you might choose one or the other
 - Why did we use CI instead of PI? How do we interpret the CI?

5.2. iPerfusion Data - Part 1

The following data is the hydrodynamic conductance (units flow rate per unit pressure [nl/min/mmHg]) for C57BL/6 mouse eyes measured using the [iPerfusion](#) system.

[5.0, 4.6, 3.5, 10.9, 7.1, 3.5, 4.3, 4.8, 3.6, 3.7, 3.9, 4.4]

You may assume that the data are sampled from a normal distribution with a standard deviation of 2 nl/min/mmHg.

- a) Find the 95% confidence interval.

[Answers: [3.8,6.1]]

- b) How would you report this result? Think about how many decimal places is reasonable.

5.3. MATLAB Simulation of CLT

This question is about gaining intuition for the CLT. I would like you to run the code I generated and consider the results. Download the m-file `Exercise5_code.m`.

You will only have to change one line of code and run it (line 5), but you may also wish (and I encourage you to) to look further into the code and how it works. Play with the numbers, see what happens!

In simple terms, for a given underlying statistical distribution (Normal, Chi-squared or F), the code

1. Generates and plots the underlying distribution (Figure 1 - top row)
2. Generates random samples using bootstrapping.
 - histograms of three random samples (Figure 1 – upper-middle row)
 - histograms of the mean values and the best fit normal distribution, with full x scale range (Figure 1 – lower-middle row) and scaled to ± 4 SE around the average xbar (Figure 1 – bottom row)
3. Analyses the data, by investigating the relationships
 - $E[\bar{X}] = \mu_{\bar{X}} = \mu_X$ (Figure 2 - left)
 - $\sigma_{\bar{X}} = \frac{\sigma_X}{\sqrt{n}}$ (Figure 2 - right)

This will generate Figures 1 and 2.

They are mostly self-explanatory, but for clarity

- The red lines in the bottom row are the PDFs of a normal distribution with the sample mean and standard deviation of the $\bar{x}$ bar values from the B samples.
- The red lines in Figure 2 are μ_X and $\frac{\sigma_X}{\sqrt{n}}$ calculated from the underlying distributions
- The green dots in Figure 2 are the averages (expected values) of sample mean (left panel) and sample SE (right panel) from the B samples.

- a) First, run the code with the normal distribution.
Line 3 of the code should read

```
Dist='Norm'
```

Set MyPath on line 10 to a location where you want to save the generated figures

Run the code (You can use the shortcut F5)

- i. Do the histograms on the middle row (the random samples) look normally distributed? Do they look similar to the underlying distribution?
- ii. Do the histograms on the bottom two rows (the means of the random samples) look normally distributed (as posited by the central limit theory)? How dependent is it on sample size? Consider if the red fit looks like the purple histogram. What do you notice about the width of the histogram? Try 'eyeballing' $\sim \pm 2D$ around the mean.
- iii. Do the data support the statement that $E[\bar{X}] = \mu_{\bar{X}} = \mu_X$?
- iv. Do the data support the idea that $\sigma_{\bar{X}} = \frac{\sigma_X}{\sqrt{n}}$? Is it dependent on sample size?

- v. What does this tell you about the robustness of CLT and the required sample size for this distribution.
- b) Second, run the code from the Chi-squared distribution, and consider the same questions
Line 3 of the code should read

```
Dist='Chi2'
```
- c) Thirdly, run the code from the F distribution, and consider the same questions

```
Dist='F'
```
- d) Comment on the following statements.
 - i. *"CLT states that as the sample size becomes large the distribution of the data approaches a normal distribution"*
 - ii. *"Sample sizes equal to or greater than 30 are often considered sufficient for the CLT to hold"*
 - iii. *"As the means were accurate for all distributions, as long as we only care about the mean, we don't need to worry"*
 - iv. *"As long as we calculate the confidence interval, then it will be ok"*

6. NHST and z-tests

6.1. Environmental Agency

An environmental agency is going to take samples from water used for cooling as it is being discharged from a power plant into a river. Regulations state that as long as the mean temperature of the discharged water is at most 60°C, there will be no negative effect on the ecosystem of the river. To investigate whether the plant is in compliance with these regulations, a number of water samples will be taken at randomly selected times and the temperature of each sample recorded. The resulting data will be used to test the hypothesis $H_0: \mu \leq 60^\circ\text{C}$. Discuss

- Whether a one- or two-tailed test is more appropriate?
- What a Type I error would mean, how likely is it to happen if $\alpha = 0.05$ and what might happen in this context as a result?
- What a Type II error would mean, how likely is it to happen if $\beta = 0.2$, and what might happen as a result?
- Considering the power of the test and your answers to (b) and (c), how would you go about selecting α for this test if you were i) the environmental agency and ii) the power plant (assuming they prioritise profit over the environment)?
- After some political discussion the key test parameters are agreed upon: a value of $\alpha = 0.05$ and a power of 90%. If the mean temperature is actually 61°C, how many samples will be required if the population standard deviation of the samples (based on prior experience) is 3°C? You can use the closest value from the statistical tables.

[Answer: 78]

- Reflections: given these parameters, is there anything else the environmental agency can do to maximise the power of the test?

6.2. Soil Specimens – Part 1

A random sample of soil specimens was obtained and measured for percentage organic matter by Mbagwu and Abeh (Prediction of engineering properties of tropical soils using intrinsic pedological parameters. *Soil Science*. 1998;163(2):93-102).

1.10	5.09	0.97	1.59	4.60	0.32	0.55	1.45	0.14	4.47
1.20	3.50	5.02	4.67	5.22	2.69	3.98	3.17	3.03	2.21
0.69	4.47	3.31	1.17	0.76	1.17	1.57	2.62	1.66	2.05

For which the sample mean is 2.48. Assume that we know the population standard deviation of 1.62.

- Should we reject the null hypothesis that the average percentage of organic matter in such soil is 3% with $\alpha=0.1$ using a z-test?

[Answers: $z = -1.758$, $z_{crit} = 1.645$]

- Find the p -value for the z-test. As the table does not have the specific value given, you can use linear interpolation: to calculate an output y corresponding to an input x , based on known points (x_0, y_0) and (x_1, y_1) , use

$$y = y_0 + (y_1 - y_0) \left(\frac{x - x_0}{x_1 - x_0} \right)$$

[Answer: $p = 0.0787$]

- c) Calculate the 95% confidence intervals on the average percent matter, and on the difference between the average percent matter and the reference mean of 3%, using the z-statistic. Report your results to two decimal places. What can you conclude from the results?

[Answers: 2.48 [1.90,3.06]%, -0.52 [-1.10,0.06]%

- d) Reflections: what considerations are important when making conclusions from this test? How certain would you be about the outcome?

6.3. Synthetic Biodiesel – Part 1

Synthetic biology techniques are being used to make yeast that can generate biodiesel. The Cetane number (CN) is a measurement of the performance of diesel fuel. Petro-diesel fuels have normally distributed CN numbers with a population average of 49 and a population SD of 12. A researcher developed an initial version of the yeast, and tested 7 samples of the biodiesel (they were aiming for 8, but one of them got contaminated). Then a small modification was made and another test was carried out, this time with 12 samples. Let X_1 and X_2 be normally distributed RVs representing the CN of the fuel generated of the initial and modified yeast. The table below contains the outcomes from both experiments.

x_1	45.5	68.6	79.6	65.9	72.4	68.7	56.4					
x_2	68.5	66.4	74.8	74.7	57.0	69.5	67.5	79.4	86.4	86.6	57.0	71.2

Carry out your calculations by hand/calculator and using statistical tables.

- a) After the first version of the yeast was tested, the researcher wanted to investigate whether the average CN of the initial biodiesel was different from petro-diesel fuels. Select and carry out a z-test with $\alpha=0.01$, under the assumption that the population standard deviation is the same. State the null and alternative hypotheses, whether we can reject the null hypothesis and what we can conclude from the result.

[Answers: $z = 3.59$, $z_{crit} = 2.575$]

- b) Estimate the p -value for this test using MATLAB and compare to α . How would you report the result? Don't worry about the CI here, although in practice you normally would.

[Answer: $p = 0.000326$]

- c) The researcher then wants to check if the modifications they made improved the performance of the biodiesel relative to the first version (if it was worse or made no difference, they would not continue with the formulation). Select and carry out an appropriate statistical test with $\alpha=0.01$. State the null hypothesis and conclude whether we can reject it (assume the same population variance between the initial and modified samples).

[Answers: $z = 1.10$, $z_{crit} = 2.33$]

6.4. iPerfusion Data – Part 2

We revisit the iPerfusion data from Exercise 5, Q2 on the hydrodynamic conductance of mouse eyes. The data are sampled from a normal distribution with a standard deviation of 2 nl/min/mmHg. Assuming that the population average is $\mu_0 = 5 \text{ nl/min/mmHg}$,

- a) How many samples would you require to detect a 40% increase in the average hydrodynamic conductance if using a two-tailed test with a power of 80% and $\alpha=0.05$.

[Answer: 8]

- b) Check your answer with the MATLAB command `sampsizepwr`. Use the help command to help you to identify the correct input parameters. Note that in the help file, the parameter $p0 = [\mu_0 \sigma_0]$.
- c) If you could only get 5 samples, what is the minimum effect size you could detect, with the specifications (using a two-tailed test with a power of 80% and $\alpha=0.05$)

[Answer: 50%]

- d) What would be the power of a test with 20 samples to detect a difference of 1 nl/min/mmHg?

[Answer: 61%]

- e) Check your answer using MATLAB `sampsizepwr`.

7. T-tests and assumptions

7.1. Soil Specimens – Part 2

We return to the random sample of soil specimens from Part 6, Question 2. You can find this data in `Data.Q1`.

1.10	5.09	0.97	1.59	4.60	0.32	0.55	1.45	0.14	4.47
1.20	3.50	5.02	4.67	5.22	2.69	3.98	3.17	3.03	2.21
0.69	4.47	3.31	1.17	0.76	1.17	1.57	2.62	1.66	2.05

For which the mean and the standard deviation of the sample are 2.481 and 1.616 respectively.

- a) Can we reject the null hypothesis that the average percentage of organic matter in such soil is 3% with $\alpha=0.1$? You can do this by hand, but you will have to use linear interpolation as the table does not have the specific value given. Check your solution with MATLAB.

[Answers: $t = -1.759$, $t_{crit} = 1.706$]

- b) Calculate the 95% confidence intervals on the average percent matter, and on the difference between the average percent matter and the reference mean of 3%, using the t -statistic (you will need linear interpolation or MATLAB). What can you conclude and how would you report the results?

[Answers: 2.48 [1.87,3.09]%, -0.52 [-1.13,0.09]%

- c) The mean and the standard deviation of the sample are 2.481 and 1.616 respectively. Can you learn anything from this?
- d) The next step is to visualise the distribution of the data using a Q-Q plot. These steps are to help you visualise the QQ plot relative to the slope of unity (the green line), which would imply a normal distribution.
- First, standardise the data (convert x to z)
 - Plot the data using the MATLAB in built function `qqplot` (or by writing your own code to do it manually)
 - Hold the plot, then plot in green the line of unity from $(x,y)=[-3 -3]$ to $(x,y)=[3 3]$
 - Use the “`axis equal`” command to make the axes occupy equal lengths on the screen
 - Set both axis limits from -3 to +3
- e) Check for normality using `adtest`, `kstest` and [swtest](#) (click the link to download the test and place in your MATLAB directory). What can you conclude from the results?

[Answer: $p_{SW} = 0.0285$, $p_{AD} = 0.0322$, $p_{KS} = 0.0019$]

- f) It could be that our data is actually sampled from a normal distribution, but that there are outliers. Check for outliers using the 1.5 IQR from the quartiles method and the z-score methods, using 2SD from the mean. If there are outliers, remove them from the data and recheck for normality. If there are not outliers, then we will proceed with the data as is.

[Answer: no outliers detected]

- g) Transformations. For each of the following transformations, create qqplots and use the tests in part (e) to establish whether the transformed variable is normally distributed.
- Lognormal distribution

[Answer: $p_{SW} = 0.0214$, $p_{AD} = 0.0743$, $p_{KS} = 0.7178$]

- reciprocal ($Y = 1/X$) – an arbitrary transformation

[Answer: $p_{SW} < 0.0001$, $p_{AD} = 0.0005$, $p_{KS} = 0.0034$]

7.2. Synthetic Biodiesel – Part 2

Synthetic biology techniques are being used to make yeast that can generate biodiesel. The Cetane number (CN) is a measurement of the performance of diesel fuel. Petro-diesel fuels normally have CN numbers in the range 47-51, with a population average of 49. A researcher developed an initial version of the yeast, and tested 7 samples of the biodiesel (they were aiming for 8, but one of them got contaminated). Then a small modification was made and another test was carried out, this time with 12 samples (as in hindsight the first sample was more variable than expected). Let X_1 and X_2 be normally distributed RVs representing the CN of the fuel generated of the initial and modified yeast. The table below contains the outcomes from both experiments

x_1	45.5	68.6	79.6	65.9	72.4	68.7	56.4					
x_2	68.5	66.4	74.8	74.7	57.0	69.5	67.5	79.4	86.4	86.6	57.0	71.2

You can find this data in `Data.Q2x1` and in `Data.Q2x2`.

Carry out your calculations by hand/calculator and using statistical tables.

- a) After the first version of the yeast was tested, the researcher wanted to investigate whether the average CN of the initial biodiesel was different from petro-diesel fuels. Select and carry out an appropriate statistical test with $\alpha=0.01$. State the null and alternative hypotheses, whether we can reject the null hypothesis and what we can conclude from the result.

[Answers: $t = 3.861$, $t_{crit} = 3.707$]

- b) Using the t-table and linear interpolation, estimate the p -value for this test and compare to α .

[Answer: $p = 0.0092$]

- c) The researcher then wants to check if the modifications they made improved the performance of the biodiesel relative to the first version. Select and carry out an appropriate statistical test with $\alpha=0.01$. State the null and alternative hypotheses, and conclude whether we can reject the null hypothesis (assume similar variance between the initial and modified samples).

[Answers: $t = 1.246$, $v = 11.13$, $t_{crit} = 3.106$]

- d) How many times larger would the difference between the average CN have to have been in order reject the null hypothesis?

[Answer: 2.5]

7.3. Happyculture vs O-BeeHive

Hydroxymethylfurfural (HMF) is a compound naturally found in low levels in honey, but enhanced by processing and long-term storage. Levels of HMF are therefore used as an indication of the quality of honey.

A marketing executive from O-BeeHive said that the average HMF levels in the honey from HappyCulture are greater than 8 mg/kg. Keen to disprove this slanderous remark, the chief happyculturist takes measurements of the HMF concentration in twenty independent samples of their honey.

$x=[5.0, 4.3, 7.6, 7.6, 2.3, 5.3, 4.6, 10.4, 16.8, 17.0, 2.3, 6.0, 1.6, 1.8, 5.5, 26.2, 2.5, 8.0, 4.4, 17.2]$;

You can find this data in `Data.Q3`. They decide to use an NSHT statistical test with $\alpha = 0.01$ to evaluate their results. A literature review indicates that HMF concentration in honey is lognormally distributed, which makes sense as it cannot be negative.

- a) Check whether their data can be considered to be normally and/or lognormally distributed.

[Answer: $p_N = 0.0019$, $p_{LN} = 0.534$]

- b) The happyculturist decides that to disprove the accusation, they will carry out a test with the null hypothesis that the average HMF concentration is greater than 8 mg/kg. Carry out the appropriate test and draw a conclusion.

[Answer: $p = 0.0425$]

- c) From the perspective of HappyCulture, this is a disappointing result. Perhaps the test was poorly formulated? What would they have concluded if they had framed the null hypothesis that the average HMF concentration is less than 8 mg/kg?

[Answer: $p = 0.9575$]

- d) Rather than pushing forward with the NHST, the happyculturist instead chooses to report the average and 99% confidence intervals as a better representation of the data. Find the 'average' and confidence interval to one decimal place

- i. Accounting for the fact that the data are sampled from a lognormal distribution

[Answer: $\bar{y}^* = 5.8 [3.5, 9.6] \text{ mg/kg}$]

- ii. Not accounting for the above (and thus implicitly assuming the data are sampled from a normal distribution)

[Answer: $\bar{y} = 7.8 [3.6, 12.0] \text{ mg/kg}$]

- e) Reflections: what have we learned?

8. Non-Parametric Tests and Regression

Data for this exercise can be found in *ExercisesPart8.mat*. This contains a single data structure with all the variables you require. Load the mat file and select variables using `Data.xxx`, where `xx` is the variable in question.

8.1. Ventilator Battery

An emergency ventilator needs to be able to run from battery power in the event of a power failure. A team submitted documentation on their model to the MHRA claiming that the median running time was at least 1.5 hours. An MHRA technician carries out tests under a range of 11 different ventilation conditions and gets the following data for run time in hours

1.5, 2.2, 0.9, 1.3, 2.0, 1.6, 1.8, 1.5, 2.0, 1.2, 1.7

- a) Use the sign-test to evaluate the hypothesis that the claim is true for $\alpha = 0.05$. Carry out your calculations by hand

[Answer: $p=0.254$]

- b) Use the sign-rank test to evaluate the hypothesis that the claim is true for $\alpha = 0.05$. Carry out your calculations by hand.

[Answer: $w^- = 15$]

- c) Now the technician gets a second ventilator. They test 7 of them and find the following data

1.3, 1.9, 0.5, 1.4, 1.2, 1.6, 1.1

Calculate the medians and use the rank-sum test to evaluate the hypothesis that they have the same median running time with $\alpha = 0.05$. Carry out your calculations by hand.

[Answer: $u = 21.5$]

8.2. Soil Specimens – Part 3

We return to the random sample of soil specimens from Part 6, Question 2 and Part 7 Question 1.

1.10	5.09	0.97	1.59	4.60	0.32	0.55	1.45	0.14	4.47
1.20	3.50	5.02	4.67	5.22	2.69	3.98	3.17	3.03	2.21
0.69	4.47	3.31	1.17	0.76	1.17	1.57	2.62	1.66	2.05

In Part 7 Question, we rejected the null hypothesis that the data are sampled from a normal distribution, and looking for outliers and transformation didn't change this. We must therefore use non-parametric tests.

Our general question of interest is should we reject the null hypothesis that the 'average' percentage of organic matter in such soil is 3% with $\alpha = 0.1$.

- a) Plot a histogram of the data minus it's median, using bins of 0.5 width (you will have to figure out an appropriate range). Do the data look to be symmetric around the median? Based on this conclusion, and those above, what statistical test is the most appropriate to use?
- b) Carry out a two-tailed sign-test on X to test the null hypothesis. Do this by hand up until the p-value, then use `binocdf` in MATLAB. Check your answer using the in-built MATLAB function `signtest`.

[Answer: $p = 0.3616$]

- c) Carry out a sign-rank test on X under the same H_0 as for part (i). This one will be a bit onerous by hand, so try it using basic MATLAB functions, as well as `tiedrank`, which automates the ranking process, then use the table to draw your conclusion. Check your value of the w -statistic using the in-built MATLAB function `signrank`, with the argument `[~,~,stats]=signrank(...`

[Answer: $w=152.5$]

- d) Use the in-built MATLAB function `signrank` with the argument `[p,~,stats]=signrank(...` using both the default method (which is used if you don't specify method) and with the argument `'method','exact'`. What do you observe?
- e) Reflections: what can we conclude from all of this analysis?

8.3. Synthetic Biodiesel – Part 3

We return to the biodiesel samples from Part 6, Question 3 and Part 7, Question 2, and repeat the analysis using a non-parametric test. We are interested in whether there is a difference between the average CN between the two values. The data can be found in Data.Q3x1 and Data.Q3x2.

- a) Use the MATLAB function `ranksum` to get the w -statistics

[Answer: $w_1 = 57, w_2 = 133$]

- b) Calculate the u -statistic, and use the table to draw a conclusion

[Answer: $u = 29$]

8.4. Crude oil imports vs chicken consumption

The table below provides data found at www.tylervigen.com/spurious-correlations on

- X – the number of barrels of crude oil imported by the US annually in billions
- Y – the number of kilograms of chicken eaten annually per person in the US

From the years 2000 to 2009

x	3.31	3.41	3.34	3.52	3.67	3.67	3.68	3.66	3.57	3.31
y	24.6	24.5	25.8	26.1	26.9	27.4	27.6	27.2	26.6	6.8

Note that I have edited the data for chicken consumption from 2009 to make the data more interesting. Let's imagine that there was a big move towards free-range chicken in 2009 for ethical reasons.

- a) Calculate the Pearson's correlation coefficient, r , and check whether r is statistically different from 0 at $\alpha = 0.05$. Do these calculations either by hand or using basic MATLAB commands, and finally confirm using `corr`.

[Answer: $r = 0.5866, p = 0.0747$]

- b) Calculate the Spearman's rank correlation coefficient, ρ_s , and check whether ρ_s is statistically different from 0 at a significance level of $\alpha = 0.05$. Do these calculations either by hand or using basic MATLAB commands, and confirm using `corr`.

[Answer: $r = 0.9329, p = < 0.0001$]

- c) Which test is more appropriate? What do the results tell you about the relationship between crude oil imports and chicken consumption in the US? What was wrong with the approach taken in this question

8.5. Heart rate vs life expectancy

The heart rate and life expectancy of a number of animals are given in the table below. The aim of this question is to evaluate whether humans fit the trend. This question can be done by hand.

	Mouse	Rat	Cat	Dog	Donkey	Elephant	Whale
Heart Rate (BPM)	500	300	150	100	50	40	30
Life Expectancy (Years)	3	10	28	32	35	55	62

- a) Let X equal life expectancy in years and Y equal the natural logarithm of the heart rate in BPM. Create a table with x and y (leave space for 2 additional rows). In the interest of efficiency, use just one decimal place on Y . Then, plot your data (by hand or with MATLAB)

- b) Calculate S_{xx} , S_{yy} and S_{xy} . For simplicity, use 3 significant figures for your calculations

[Answer: $S_{xx} = 2780$, $S_{yy} = 6.66$, $S_{xy} = -131$]

- c) Calculate estimators for b_1 and b_0

[Answer: $\hat{b}_1 = -0.0471$, $\hat{b}_0 = 6.15$]

- d) Calculate the values of $\hat{y}_i$ for the x_i values, and hence calculate the residuals (using 3 sf). Add these to the table.

- e) Calculate s_ε^2 .

[Answer: $s_\varepsilon^2 = 0.0952$]

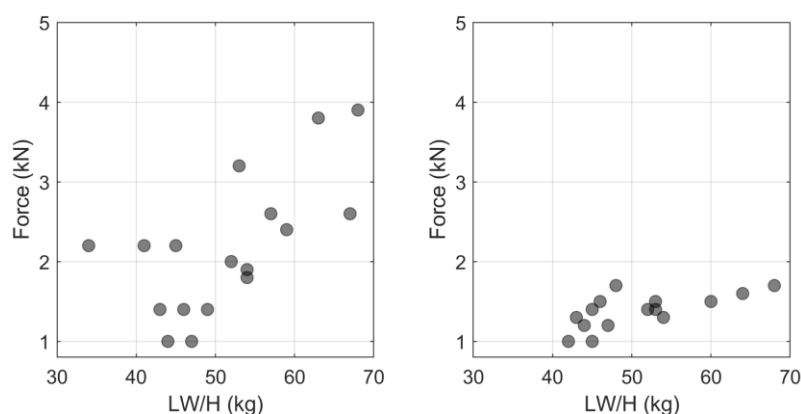
- f) Humans have a life expectancy of 70 years. Using the 95% Prediction Bands, estimate a range for the heart rate of a given human in BPM.

[Answer: $PI_{HR} = [6.2, 47.9]$]

- g) The actual heart rate of humans is approximately 60 BPM. Is it appropriate to conclude that humans 'fit the trend' or 'are an outlier'? What assumptions are involved in this conclusion? Can you produce a figure that visualises this analysis?

8.6. Hip and knee forces

For a prosthesis design, we need to know the forces that a joint undergoes as the person moves. The figure below shows measurements from Paul (1970) of hip (left panel) and knee (right panel) forces against WL/H (L is stride length, W is weight and H is height).



- a) Calculate the Pearson's correlation coefficient, coefficient of determination and corresponding p-value for both cases. You can do this using the built-in MATLAB function or by writing your own code (or ideally, using both approaches).

[Answer: Hips $r_h = 0.6503$, $p_h = 0.0047$, $r_h^2 = 0.4229$,
Knees $r_k = 0.6679$, $p_k = 0.0065$, $r_k^2 = 0.4461$]

- b) What can you conclude from these statistics? What additional information is required?
- c) Calculate the slope and the 95% confidence interval on the slope for both knee and hip data. What can you conclude from these results about whether the force on the hips or knees is more sensitive to WL/H?

[Answer: Hips 0.061 [0.022,0.099], Knees 0.018 [0.006,0.031]]

9. Multiple Comparisons

9.1. E. coli doubling rate

A colony of Escherichia coli bacteria can double every 20 minutes under ideal laboratory conditions. A study assessed the doubling times when different nutrients were supplied. To this end, 50 colonies were divided into five groups of ten colonies. Each group was given a different nutrient, and the doubling times were measured for each colony. The researchers obtained the following data:

Nutrient	1	2	3	4	5
Sample mean	10.5	14.8	15.7	16.0	21.6
Sample standard deviation	4.5	6.8	6.5	6.7	6.0

State the null hypothesis and determine, with $\alpha=0.05$, whether the null hypothesis can be rejected using the F table.

[Answer: $F=4.136$, reject]

9.2. Titanium bone implants: parametric

This question is inspired by [Niinomi and Nakai](#) (2011). Titanium alloys for bone implants are required to have a low Young's modulus in order to enhance bone remodelling and inhibit bone atrophy. Processing methods such as Rolling and Swaging can be used to lower the Young's Modulus.

Researchers wanted to investigate which of four methods (two rolling processes and two swaging processes – labelled A, B, C, D) would yield the lowest Young's Modulus. 7 samples were collected for each process, but some sample flaws led to failure of a few of the tests which had to be discarded. The results of the experiments (Young's modulus in GPa) are provided below. The grand mean and grand variance are 65.59 and 38.43 respectively.

								$\bar{x}_i$	s_i
A	57.1	54.2	66.1	66.0	64.5	66.5	69.4	63.40	5.56
B	64.9	57.2	64.0	60.9	57.5	54.0	63.2	60.24	4.10
C	70.2	72.2	68.0	64.2	72.9	71.7	Fail	69.87	3.28
D	76.0	74.0	75.3	66.3	65.3	63.7	Fail	70.10	5.58

Use with $\alpha=0.05$ and round to 2 decimal places.

- Identify and define I , J_i and N
- Calculate SS_T with the most efficient option from the formula sheet

[Answer: $SS_T=960.75$]

- Calculate SS_M

[Answer: $SS_M = 465.88$]

- Calculate SS_R with the most efficient option from the formula sheet

[Answer: $SS_R = 495.15$]

- e) Check that the calculated values of SS_T , SS_M and SS_R are consistent with each other. What can we learn based on the relative sizes of the SS terms?
- f) Calculate the degrees of freedom and mean sums of squares. What can we learn based on the relative sizes of the MS terms?

[Answer: $MS_M = 155.29$, $MS_R = 22.51$]

- g) Calculate the F-statistic and hence using the table identify whether the null hypothesis can be rejected.

[Answer: $F = 6.90$, reject]

- h) Use the Bonferroni approach to identify which groups come from populations that have different means with $\alpha = 0.05$. I have provided the t-test outcomes below to save you the calculations.

Group 1	Group 2	$\bar{y}$	$s_{\bar{y}}$	t	ν	p
A	B	3.16	2.61	1.21	11.0	0.25
A	C	-6.47	2.49	-2.60	9.9	0.027
A	D	-6.70	3.10	-2.16	10.7	0.054
B	C	-9.63	2.05	-4.69	11.0	0.00065
B	D	-9.86	2.76	-3.57	9.1	0.0059
C	D	-0.23	2.64	-0.09	8.1	0.93

- i) Report your results relative to the initial aim of the study. This is a chance for you to think about how to interpret the data as a whole not just the outputs from the ANOVA.

9.3. Titanium bone implants: non-parametric

In this question, we will process the data from Question 2 using non-parametric statistics. To save you the pain, here are the ranks, with the colour coding **A, B, C, D**.

1	2	3	4	5	6	7	8	9	10	11	12	13
54.0	54.2	57.1	57.2	57.5	60.9	63.2	63.7	64.0	64.2	64.5	64.9	65.3
14	15	16	17	18	19	20	21	22	23	24	25	26
66.0	66.1	66.3	66.5	68.0	69.4	70.2	71.7	72.2	72.9	74.0	75.3	76.0

- a) Find the sum ranks and sample sizes for each group and hence calculate the H statistic

[Answer: $h = 12.5$]

- b) Calculate the degrees of freedom and hence find the critical χ^2 statistic for $\alpha = 0.05$. What can you conclude?

9.4. Intraocular pressure medication

A new medication has been proposed that might lower intraocular pressure and thus help reduce progression of glaucoma. The medication has been tested on cells and ex vivo tissue, and has shown an effect that is expected to result in lowering of the intraocular pressure. The next step is to do a small animal study. Intraocular pressures in 6 mice were measured before the first dose of treatment was administered (baseline) and again at 2, 4, and 12 weeks after the initiation of treatment. The aim of the study is to ascertain whether the drug lowers intraocular pressure and if so, for how long.

In this context, the multiple time-points correspond to repeated measurements on the same individual. Each time point is a 'group' to be considered. We can do using a repeated-measures ANOVA. The data on intraocular pressure is provided below

	Baseline	Week 2	Week 4	Week 12	$\bar{X}_j$
	15	14	12	14	13.8
	17	18	10	12	14.3
	13	13	9	10	11.3
	14	12	13	14	13.3
	12	8	10	10	10.0
	16	16	14	11	14.3
$\bar{X}_i$	14.5	13.5	11.3	11.8	$\bar{X} = 12.8$

State the null hypothesis and determine whether the null hypothesis can be rejected with $\alpha=0.05$.

[Answer: $F = 3.8$, reject]

9.5. Corrosion fatigue

Corrosion fatigue in metals has been defined as the simultaneous action of cyclic stress and chemical attack on a metal structure. A widely used technique for minimising corrosion fatigue damage in aluminium involves the application of a protective coating. Samples were created with one of three types of surface coating:

1. Uncoated: no coating
2. Anodized: sulfuric acid anodic oxide coating
3. Conversion: chromate chemical conversion coating

And were subjected to cyclic stress under one of 3 levels of humidity

1. Low: 20-25% relative humidity
2. Medium: 55-60% relative humidity
3. High: 86-91% relative humidity

The number cycles to failure, expressed in 1000's are given in the table

	Low			Medium			High		
	U	A	C	U	A	C	U	A	C
	361	114	130	314	322	252	1344	78	586
	469	1032	1482	522	471	874	1216	466	524
	466	1236	841	244	306	405	1027	387	402
	937	92	539	739	30	755	1097	107	751
	1069	533	1595	261	68	847	1011	130	846
	1357	211	754	134	398	573	1011	327	529
$\bar{X}_{gi}$	777	536	890	369	266	618	1118	249	606

The sum square for the residuals is $SS_R = 4,682,363$ and $SS_M = 4,054,529$. You can round to the nearest integer (except for your F-statistic) for this question to simplify your calculations (this will affect individual values, but not overall conclusions).

State the null hypotheses and determine, at $\alpha=0.05$, whether the null hypotheses can be rejected. Completing the table of means below might be helpful. For F-statistics, try to reach a conclusion using the tables, but without interpolation.

	Low	Medium	High	$\bar{X}_t$
Uncoated				
Anodised				
Conversion				
$\bar{X}_g$				$\bar{X} =$

Hence complete the ANOVA table.

Source	SS	ν	MS	F	F_{crit}	Reject H_0 ?
Humidity					3.20	
Coating					3.20	
Interaction					2.58	
Error						
Total						

Bonus - ANOVA with MATLAB

9.6. Parametric titanium bone implants with MATLAB

For the data on titanium alloys (A,B,C,D)

- Check whether normality is rejected for any of the groups
- Hence choose the appropriate test for homoscedasticity and check if it is rejected.
- Carry out the appropriate test to evaluate if any of the averages of groups A, B, C and D are different from each other
- Carry out a post-hoc test using Tukey's HSD test and draw conclusions with $\alpha=0.05$.

9.7. Non-parametric titanium bone implants with MATLAB

We now also consider a 5th process, E.

- Carry out the necessary checks to see if the assumptions required for ANOVA are met
- Carry out the appropriate test to evaluate if any of the averages of groups A, B, C, D and E are different from each other

9.8. Intraocular pressure medication with MATLAB

For the IOP data, carry out a repeated measures ANOVA using `anova_rm` (which needs to be downloaded). What can you conclude with $\alpha=0.05$? The data is in the variable IOP.

9.9. Corrosion fatigue with MATLAB

For the cyclic stress experiments and the factors coating and humidity, carry out a 2-way ANOVA